

Azure Setup Requirements

Database tables, API request and response bodies, email services, and the order to build them in for the Canadian Islamic Union website.

Canadian Islamic Union

ciucanada.ca | August 2026

1. Recommended Azure stack

Use Azure SQL Database plus a small REST API. The CIU website already expects that shape: when `AZURE_API_BASE_URL` is set, Next.js calls Azure and uses the JSON as-is. Do not wrap responses like `{ data: [...] }`. Arrays for lists, objects for creates, and `{ error: "message" }` only on failure.

Piece	Use
Database	Azure SQL Database (one database, e.g. ciu)
API	Azure Functions or App Service at <code>AZURE_API_BASE_URL</code>
API auth	<code>AZURE_API_KEY</code> as Authorization: Bearer and/or x-api-key
Transactional email	Azure Communication Services Email (or SendGrid)
Mailing list later	Same ACS/SendGrid, or Mailchimp / Constant Contact

IDs are UUID strings. Dates are ISO-8601 UTC, e.g. 2026-08-17T19:40:00.000Z. DELETE endpoints return HTTP 204 with an empty body.

2. How the website talks to Azure

Set these environment variables on the website. `lib/portal/azure-store.ts` already maps to the paths below. Until `AZURE_API_BASE_URL` is set, the site uses an in-memory store that resets on deploy.

Variable	Purpose
<code>AZURE_API_BASE_URL</code>	API origin with no trailing slash
<code>AZURE_API_KEY</code>	Shared secret sent as Bearer token and x-api-key
<code>ADMIN_PASSWORD</code>	Temporary shared password on Next.js until <code>/auth/login</code> uses users

Paths already called by the website

Method	Path	Used by
GET	<code>/announcements</code>	Public ticker + admin
POST	<code>/announcements</code>	Admin create
PATCH / DELETE	<code>/announcements/{id}</code>	Admin update / delete
GET / POST	<code>/events</code>	Public Events + admin
PATCH / DELETE	<code>/events/{id}</code>	Admin update / delete
GET / POST	<code>/contacts</code>	Contact form + admin inbox
GET / POST	<code>/registrations</code>	<code>/register</code> + admin
GET / POST	<code>/history</code>	Admin activity log
POST	<code>/auth/login</code>	Staff sign-in (add this)
GET / POST	<code>/users</code>	Staff accounts (add this)

Method	Path	Used by
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3. Required tables

Announcements, events, contacts, registrations, and history are already used by the website. Add users as well so every change is tied to who signed in. Today the portal only stores a typed email cookie, so history can say unknown. users is the real staff account table.

3.1 users (staff who authenticate)

This is not the public membership list. users are people who log into /admin. Store a password hash only, never the raw password. Never return passwordHash in any response. role is admin or staff. Inactive users must not be able to sign in.

Column	Type	Notes
id	uniqueidentifier	PK, generated by Azure
email	nvarchar(256)	Required, unique, lowercase
displayName	nvarchar(200)	Required. Shown in the portal
role	nvarchar(20)	admin staff
passwordHash	nvarchar(256)	Hash only. Never send this back
isActive	bit	Default 1. 0 = cannot sign in
lastLoginAt	datetime2	Null until first successful login
createdAt	datetime2	ISO string in JSON
updatedAt	datetime2	ISO string in JSON

POST /auth/login - body

```
{
  "email": "staff@ciucanada.ca",
  "password": "the-staff-password"
}
```

Login response 200 - never include passwordHash

```
{
  "id": "11111111-1111-1111-1111-111111111111",
  "email": "staff@ciucanada.ca",
  "displayName": "Amina Khan",
  "role": "admin",
  "isActive": true,
  "lastLoginAt": "2026-08-17T19:40:00.000Z",
  "createdAt": "2026-08-01T12:00:00.000Z",
  "updatedAt": "2026-08-17T19:40:00.000Z"
}
```

POST /users - create staff - body

```
{
  "email": "staff@ciucanada.ca",
  "displayName": "Amina Khan",
  "role": "admin",
  "password": "temporary-password",
  "isActive": true
}
```

Azure hashes password into passwordHash and returns the same public user object as login. GET /users returns that array. PATCH /users/{id} can change displayName, role, isActive, or password. Do not hard-delete; set isActive to false. Failed login: HTTP 401 with { error: "Incorrect email or password." }.

members = public mailing list. users = staff portal accounts. Do not mix them. After login, history rows must store userId plus email so you can see who created or edited a record.

3.2 announcements

Public site ticker. Admin can create, edit, hide, and delete. Public GET returns active items only.

Column	Type	Notes
id	uniqueidentifier	PK, generated by Azure
message	nvarchar(500)	Required
href	nvarchar(500)	Optional. /Events or a full URL
active	bit	Default 1
createdAt	datetime2	ISO string in JSON
updatedAt	datetime2	ISO string in JSON

POST /announcements - body

```
{
  "message": "Eid prayer this Friday",
  "href": "/Events",
  "active": true
}
```

Response 201: full row including id, createdAt, updatedAt. PATCH accepts any subset of message, href, active.

3.3 events

Public /Events list. Admin CRUD. Category must be one of: education, youth, family, community, spiritual, volunteer.

Column	Type	Notes
id	uniqueidentifier	PK
title	nvarchar(200)	Required
category	nvarchar(32)	See allowed values above
dateLabel	nvarchar(120)	Display text, e.g. Every Tuesday
date	date	Optional sort date YYYY-MM-DD
time	nvarchar(80)	Required

Column	Type	Notes
description	nvarchar(max)	Required
tags	nvarchar(max)	JSON array of strings
href	nvarchar(500)	Required button link
buttonLabel	nvarchar(80)	Default "View Details"
image	nvarchar(1000)	Optional image URL
recurring	bit	Default 0
featured	bit	Default 0
createdAt / updatedAt	datetime2	ISO strings in JSON

POST /events - body

```
{
  "title": "Weekly Quran Class",
  "category": "education",
  "dateLabel": "Every Tuesday & Thursday",
  "date": "2026-09-01",
  "time": "7:00 PM",
  "location": "Canadian Islamic Centre",
  "description": "Open to adults and all learners.",
  "tags": ["quran", "education"],
  "href": "/Education/ciu",
  "buttonLabel": "View Details",
  "image": "https://...",
  "recurring": true,
  "featured": false
}
```

Response: full event object. GET /events returns an array, newest date first.

3.4 contacts

Public contact form into the admin inbox. No update or delete in the app today.

Column	Type	Notes
id	uniqueidentifier	PK
firstName	nvarchar(100)	Required
surname	nvarchar(100)	Required
name	nvarchar(200)	Azure should set "First Last"
email	nvarchar(256)	Required
phone	nvarchar(40)	Required
subject	nvarchar(200)	Required
message	nvarchar(max)	Required

Column	Type	Notes
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POST /contacts - body (what the site sends)

```
{
  "firstName": "Amina",
  "surname": "Khan",
  "email": "amina@example.com",
  "phone": "905-555-0123",
  "subject": "Weekend school",
  "message": "Do you still have space in Grade 4?"
}
```

Response 201: include generated id, name, and createdAt. GET /contacts: array, newest first.

3.5 registrations

One table for Quran class and kids weekend school. program is quran or kids. Kids rows require studentAge, grade (1-12), and parentName. Quran rows leave those empty.

Column	Type	Notes
id	uniqueidentifier	PK
program	nvarchar(16)	quran or kids
studentName	nvarchar(200)	Required (learner or child)
studentAge	nvarchar(20)	Required for kids; empty for quran
grade	nvarchar(8)	Kids only: "1" through "12"
parentName	nvarchar(200)	Kids only
email	nvarchar(256)	Required
phone	nvarchar(40)	Required
notes	nvarchar(max)	Optional
createdAt	datetime2	ISO string in JSON

Quran - POST /registrations

```
{
  "program": "quran",
  "studentName": "Yusuf Ali",
  "email": "yusuf@example.com",
  "phone": "416-555-0199",
  "notes": "Beginner"
}
```

Kids - POST /registrations

```
{
  "program": "kids",
  "studentName": "Fatima Ali",
  "studentAge": "9",
}
```

```

"grade": "4",
"parentName": "Sara Ali",
"email": "sara@example.com",
"phone": "416-555-0199",
"notes": ""
}

```

Response 201: full row. GET /registrations returns all programs; the admin UI filters by tab.

3.6 history

Staff audit log. Every create, update, delete, sign-in, and sign-out writes a row. Tie it to users: store userId from the signed-in account, and keep adminEmail as a readable copy of users.email. If Azure only receives userId, look up the email from users.

Column	Type	Notes
id	uniqueidentifier	PK
userId	uniqueidentifier	FK to users.id. Null only for old unknown rows
adminEmail	nvarchar(256)	Copy of users.email at the time of the action
action	nvarchar(32)	authenticated signed-out created updated deleted
area	nvarchar(32)	session users announcements events contacts registrations
summary	nvarchar(500)	Human-readable sentence
entityId	uniqueidentifier	Optional related row
createdAt	datetime2	ISO string in JSON

POST /history - body

```

{
  "userId": "11111111-1111-1111-1111-111111111111",
  "adminEmail": "staff@ciucanada.ca",
  "action": "created",
  "area": "events",
  "summary": "Added event: Weekly Quran Class",
  "entityId": "22222222-2222-2222-2222-222222222222"
}

```

4. Tables not wired yet

These public forms exist on the site but currently show a coming-soon toast. Create the tables now so the website can POST later.

4.1 members (mailing list)

From /Membership. This is the community email list. emailTopics values: events, education, family-programs, community-news, charitable-projects. membershipType is individual or family.

Column	Type	Notes
id	uniqueidentifier	PK

Column	Type	Notes
fullName	nvarchar(200)	Required
email	nvarchar(256)	Required, unique
phone	nvarchar(40)	Optional
city	nvarchar(100)	Optional
membershipType	nvarchar(20)	individual family
emailTopics	nvarchar(max)	JSON array of topic keys
notes	nvarchar(max)	Optional message field
agreedToEmails	bit	Must be 1
unsubscribedAt	datetime2	Null until they opt out
source	nvarchar(40)	membership events-newsletter
createdAt	datetime2	ISO string in JSON

POST body to plan for

```
{
  "fullName": "Amina Khan",
  "email": "amina@example.com",
  "phone": "905-555-0123",
  "city": "Mississauga",
  "membershipType": "family",
  "emailTopics": ["events", "education"],
  "message": "",
  "agreement": true
}
```

4.2 volunteers

From /Services/volunteer. This is staff follow-up, not a newsletter.

ageGroup: high-school | adult | senior. availability: weekdays | weekends | evenings | flexible. roles: event-coordination, community-engagement, education-programs, masjid-facility, outreach, admin-support, student-mentor.

Column	Type	Notes
id	uniqueidentifier	PK
fullName	nvarchar(200)	Required
email	nvarchar(256)	Required
phone	nvarchar(40)	Required
ageGroup	nvarchar(20)	high-school adult senior
roles	nvarchar(max)	JSON array of role keys
availability	nvarchar(20)	weekdays weekends evenings flexible
volunteerHours	nvarchar(8)	yes no empty

Column	Type	Notes
agreement	bit	Required
createdAt	datetime2	ISO string in JSON

POST body to plan for

```
{
  "fullName": "Omar Hassan",
  "email": "omar@example.com",
  "phone": "905-555-0144",
  "ageGroup": "adult",
  "roles": ["event-coordination", "masjid-facility"],
  "availability": "weekends",
  "volunteerHours": "no",
  "message": "Happy to help with setup.",
  "agreement": true
}
```

4.3 Event newsletter

The Events page form collects fullName, email, and consent. Do not create a second mailing list. Store these in members with source = events-newsletter.

```
{ "fullName": "Amina Khan", "email": "amina@example.com" }
```

5. What does not need a table

Feature	Why
Donate / e-transfer / PayPal	Copy-paste Canadianislamicunion@gmail.com only
Funeral, counseling, matrimonial	Point people to the Contact form
Prayer times, About, Media	Static content on the website
Shared ADMIN_PASSWORD cookie	Temporary only. Replace with users + /auth/login

6. Email services

You need a real email product for notifications and mailing lists. Azure Communication Services Email is the Azure-native option. From-address should be noreply@ciucanada.ca or info@ciucanada.ca after DNS (SPF/DKIM).

Must have email

Trigger	Send to	Why
Contact form	info@ciucanada.ca	Staff alert with the message. Optional auto-reply to sender.
Class registration	Staff inbox	New Quran/kids signup. Optional confirmation to the family.
Volunteer form	Staff inbox	Team will follow up. Optional confirmation to the volunteer.

Trigger	Send to	Why
Membership signup	Members / volunteers	ACS, contact, contact emails, contact emails, contact emails
Events newsletter	Same list engine	Merge into members unless you keep a separate list.
Staff invite / reset	That user's email	Optional. Send a temporary password when a users row is created.

Does not need email sending

Item	Notes
Donate / PayPal / e-transfer	Display and copy Canadianislamicunion@gmail.com
Footer / Contact page email	mailto: links only
Admin history	Database only
Announcements / events on the site	Shown on the website, not emailed unless you later blast the list

Suggested mail types

- Transactional (ACS): contact alert, registration received, volunteer received.
- Marketing / newsletter (ACS or Mailchimp): membership and event updates, topic tags, unsubscribe.
- Canada CASL: store agreedToEmails plus timestamp. Every marketing mail needs unsubscribe.

Staff notify address: info@ciucanada.ca. Donations stay on Canadianislamicunion@gmail.com and must not mix with the mailing list.

7. Build order

Step	Work
1	Azure SQL + users first, then announcements, events, contacts, registrations, history
2	Add POST /auth/login and store userId on every history row
3	Point the website at AZURE_API_BASE_URL and AZURE_API_KEY
4	ACS Email for contact and registration alerts
5	Create members and volunteers tables, then connect those forms
6	Merge the events newsletter into members (source field)
7	Optional: admin screens for members/volunteers, and announcement emails to the list

8. Error and success contract

- Success lists: a JSON array of objects. No wrapper.
- Success create: the saved object with id and timestamps. HTTP 201.
- Success update: the saved object. HTTP 200.
- Success delete: HTTP 204, empty body.
- Failure: { "error": "Human readable message" } with HTTP 400/401/404/500.

```
{ "error": "Name, email, and phone are required." }
```